SFM-V1.7 Fingerprint Module

Communication Protocol Overview

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Communication Protocol Overview

The host and fingerprint module communicate via CMD and ACK packets beginning and ending with F5. The host sends 8-byte CMD commands and data packets (usually more than 8 bytes); the module responds with 8-byte ACK commands and data packets (usually more than 8 bytes).

1 Eight-byte format

Data format:

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	TYPE	P1	P2	P3	0	CHK	0xF5
ACK	0xF5	TYPE	Q1	Q2	Q3	0	CHK	0xF5

Notes:

TYPE: CMD/ACK type

P1, P2, P3: CMD parameters

Q1, Q2, Q3: ACK parameters

Q3 always indicates the command execution result:

```
#define ACK_SUCCESS          0x00    // Executed successfully
#define ACK_FAIL             0x01    // Execution failed
#define ACK_FULL             0x04    // Database full
#define ACK_NOUSER           0x05    // User not found
#define ACK_USER_EXIST       0x07  // User already exists
#define ACK_TIMEOUT          0x08    // Image acquisition timed out
#define ACK_HARDWAREERROR    0x0A  // Hardware error
#define ACK_IMAGEERROR       0x10    // Image error
#define ACK_BREAK            0x18    // Terminate current command
#define ACK_ALGORITHMFAIL 0x11 // Film spoofing detection
#define ACK_HOMOLOGYFAIL 0x12 // Same-finger verification error
```

CHK: XOR checksum of bytes 2 through 6.

CHK XOR checksum

```
                    ;i++)
{
 CheckXOR ^= TxBuf[i];
}
```

2 Packets longer than eight bytes: packet header + data packet

Packet header format:

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	TYPE	Hi(Len)	Low(Len)	0	0	CHK	0xF5
ACK	0xF5	TYPE	Hi(Len)	Low(Len)	Q3	0	CHK	0xF5

Notes:

Q3 is defined in the preceding section.

Len: two-byte length of the Real Data in the data packet.

Hi(Len): high byte of actual data length.

Low(Len): low byte of actual data length.

CHK: XOR checksum of bytes 2 through 6.

Data packet format:

Byte	1	2...Len+1	Len+2	Len+3
CMD	0xF5	Real Data	CHK	0xF5
ACK	0xF5	Real Data	CHK	0xF5

Notes:

Len ： Length of Real Data.

CHK: XOR checksum of bytes 2 through Len+1. The data packet is transmitted immediately after its packet header.

3 3 Command Definitions

3.1 3.1 User Enrollment Overview

Two major enrollment methods are available: 3C3R and NCNR (standard default: 3C3R).

These modes are described below:

1. 3C3R uses multiple commands and responses, providing a practical balance between user operating complexity and flexibility.

2. NCNR permits an application-specific number of enrollment captures; as with 3C3R, N=3 generally provides the best user enrollment experience.

The module provides film spoofing detection by default. Do not hold the finger down continuously during enrollment. Use 3C3R (or NCNR for multiple captures); lift the finger.

During any enrollment process, ACK TYPE 0x03 indicates completion. When ACK Q3 is 0, enrollment succeeded and ACK Q1/Q2 give the user's ID (high/low bytes).

Q1 and Q2 are the high and low 8 bits of the user ID. Details follow.

3.1.1 Multi-press Enrollment (3C3R) (8-byte CMD/ACK)

First enrollment command and response

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x01	UserID(Higher8bits)	UserID(Lower8 bits)	User Role (1/2/3)	0	CHK	0xF5
ACK	0xF5	0x01	UserID(Higher8bits) 0	UserID(Lower8 bits) 0	ACK_SUCCESS ACK_FAIL ACK_FULL ACK_TIMEOUT ACK_BREAK ACK_IMAGEFAIL ACK_HARDWAREERROR ACK_IMAGEERROR	0	CHK	0xF5

Set the user ID and role in the first command. User ID range: 1-10000. If set to 0, the module allocates an unused ID automatically, subject to its maximum user capacity.

Second enrollment command and response

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x02	0	0	0	0	CHK	0xF5
ACK	0xF5	0x02 0x03	0	0	ACK_SUCCESS ACK_FAIL ACK_FULL ACK_TIMEOUT ACK_BREAK ACK_IMAGEFAIL	0	CHK	0xF5

If ACK byte 2 is 0x03, enrollment terminated early. Byte 5 indicates the cause; the user must restart enrollment.

Third enrollment command and response

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x03	0	0	0	0	CHK	0xF5
ACK	0xF5	0x03	UserID(Higher8bits) 0	UserID(Lower8 bits) 0	ACK_SUCCESS ACK_FAIL ACK_FULL ACK_TIMEOUT ACK_BREAK ACK_IMAGEFAIL ACK_ALGORITHMFAIL ACK_HOMOLOGYFAIL	0	CHK	0xF5

If any of the three enrollment attempts fails, enrollment must restart from the beginning.

For example, if the third attempt does not return ACK_SUCCESS, repeat enrollment from the first command. ACK bytes 3 and 4 indicate the user ID only upon success.

3.1.2 Multi-press Enrollment (NCNR) (8-byte CMD/ACK)

After configuring NCNR, issue N enrollment commands.

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x01	UserID(Higher8bits)	UserID(Lower8 bits)	ROLE	0	CHK	0xF5
…	0xF5	0x01	0	0	0	0	CHK	0xF5
…	0xF5	0x01	0	0	0	0	CHK	0xF5
ACK	0xF5	0x03	UserID(Higher8bits)	UserID(Lower8 bits)	ACK_SUCCESS ACK_FAIL ACK_FULL ACK_TIMEOUT ACK_BREAK ACK_IMAGEFAIL ACK_ALGORITHMFAIL ACK_HOMOLOGYFAIL	0	CHK	0xF5

For N>3, send F5 01 00 00 00 00 01 F5 repeatedly. Before the final attempt, responses are F5 01 00 00 00 00 01 F5. On the final response, examine success/failure, or TYPE=03, to determine the result.

3.2 3.2 Delete Specific User / Users with a Given Role (8-byte CMD/ACK)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x04	User ID(Higher 8 Bits)	User ID(Low 8 Bits)	Role	0	CHK	0xF5
ACK	0xF5	0x04	0	0	ACK_SUCCESS ACK_NOUSER	0	CHK	0xF5

Description:

Delete a fingerprint by user ID, or delete all users belonging to a specified role.

Example 1:

Delete user 0001: send the specified-ID deletion command via the serial port.

F5 04 00 01 00 00 05 F5

The module searches its fingerprint database for user 0001. If found, it deletes the fingerprint and returns via the serial port:

F5 04 00 00 00 00 04 F5 (deletion successful)

If not found, it returns an error:

F5 04 00 00 05 00 01 F5 (user does not exist)

Example 2:

Delete all users whose role is 02.

Send: F5 04 00 00 02 00 06 F5

Return: F5 04 00 00 00 00 04 F5 (deletion successful)

Example 3:

Delete user 00 0A and all users with role 03.

Send: F5 04 00 0A 03 00 0D F5

Return: F5 04 00 00 00 00 04 F5 (deletion successful)

3.3 3.3 Delete All Users (8-byte CMD/ACK)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x05	0	0	0	0	CHK	0xF5
ACK	0xF5	0x05	0	0	ACK_SUCCESS	0	CHK	0xF5

Description:

Delete all fingerprints in the database.

Example:

Send the delete-all-users command via the serial port.

F5 05 00 00 00 00 05 F5

Success response: F5 05 00 00 00 00 05 F5

Wait for the module's response before issuing any other command.

On receipt, the module deletes all saved fingerprints. This command does not return failure unless it interrupts another command.

For example:

If sent during enrollment,

F5 05 00 00 00 00 05 F5

enrollment stops and the delete-all command is not executed.

3.4 3.4 Get Total User Count (8-byte CMD/ACK)

Bytes	1	2	3	4	5	6	7	8
CMD	0xF5	0x09	0	0	0	0	CHK	0xF5
ACK	0xF5	0x09	Usercount(higher8bits)	Usercount(lower8bits)	ACK_SUCCESS	0	CHK	0xF5

Description:

Return the number of enrolled users in the fingerprint database.

Example: send the get-user-count command via the serial port.

F5 09 00 00 00 00 09 F5

The module counts the enrolled users and returns the number through the serial port.

F5 09 00 01 00 00 08 F5

This indicates 0001 user in the fingerprint database.

3.5 3.5 Get User Role (8-byte CMD/ACK)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x0A	UserID (Higher 8bits)	UserID (Low 8Bits)	0	0	CHK	0xF5
ACK	0xF5	0x0A	0	0	User role(1/2/3) ACK_NOUSER	0	CHK	0xF5

Description:

Check whether a user ID is already assigned in the fingerprint database.

Example: request the role of user 0001 via the serial port.

F5 0A 00 01 00 00 0b F5

If user 0001 does not exist in the database, the module

returns:

F5 0A 00 00 05 00 0F F5

If the user exists, it returns:

F5 0A 00 00 01 00 0B F5

3.6 3.6 1:1 Matching (8-byte CMD/ACK)

Bytes	1	2	3	4	5	6	7	8
CMD	0xF5	0x0B	UserID (higher 8 bits)	UserID (lower 8 bits)	0	0	CHK	0xF5
ACK	0xF5	0x0B	0	0	ACK_SUCCESS ACK_FAIL ACK_TIMEOUT ACK_BREAK ACK_HARDWAREERROR ACK_IMAGEERROR	0	CHK	0xF5

Description:

Compare a captured fingerprint with the enrolled fingerprint for a specified user ID.

Example:

Send the 1:1 matching command for user 0001 via the serial port.

F5 0B 00 01 00 00 0A F5

The module searches for user 0001 in the fingerprint database.

If absent, it returns immediately:

F5 0B 00 00 05 00 0E F5

If present, the module waits for a fingerprint. If no finger is pressed within 8 seconds:

F5 0B 00 00 08 00 03 F5

If the fingerprint is captured but does not match, the serial port returns:

F5 0B 00 00 01 00 0A F5

If matching succeeds, it returns:

F5 0B 00 00 00 00 0B F5

If another command arrives during capture, the current matching command is interrupted;

the interrupting command is not executed. The module returns:

F5 0B 00 00 18 00 13 F5

3.7 3.7 1:N Matching and Matching by User Role (8-byte CMD/ACK)

Bytes	1	2	3	4	5	6	7	8
CMD	0xF5	0x0C	0	0	Role	0	CHK	0xF5
ACK	0xF5	0x0C	UserID (higher8 bits)	UserID (lower 8 bits)	User role(1/2/3)	0	CHK	0xF5

Description:

Compare a captured fingerprint against all users, or only users of a specified role.

On success, return the first matching user ID and role; otherwise the ID is 00 00.

Example 1:

Send the following command via the serial port:

F5 0C 00 00 00 00 0C F5

The module waits for the user to present a fingerprint.

If no fingerprint is presented, it times out and returns a timeout error:

F5 0C 00 00 08 00 04 F5

If a fingerprint is presented but does not match any enrolled fingerprint, return:

F5 0C 00 00 00 00 0C F5

If matching succeeds, return the first matching user's ID：

F5 0C 2bytesID 1byteRole 00 CHK F5

(e.g.: F5 0C 00 05 01 00 08 F5)

Example 2:

Match fingerprints for role 02 (user ID 00 0A, role 02).

Send: F5 0C 00 00 02 00 0E F5. Only fingerprints with role 02 are checked.

Return: F5 0C 00 0A 02 00 04 F5 (matching ID and role on success)

Return: F5 0C 00 00 02 00 0E F5 (ID 00 00 and role on failure)

Software can determine success by checking whether 2bytesID is nonzero.

2bytesID!=0.

3.8 3.8 Get an Unused User ID (8-byte CMD/ACK)

Bytes	1	2	3	4	5	6	7	8
CMD	0xF5	0x0D	0	0	0	0	CHK	0xF5
ACK	0xF5	0x0D	UserID (higher 8 bits)	UserID (lower 8 bits)	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

Description: The module returns the first available user ID.

Example:

Send via the serial port:

F5 0D 00 00 00 00 0D F5

The module responds:

F5 0D 00 02 00 00 0F F5

This means user ID 0002 is available.

3.9 3.9 Set Serial Port Baud Rate (8-byte CMD/ACK)

Bytes	1	2	3	4	5	6	7	8
CMD	0xF5	0x21	0	New baud rate ID	Flag	0	CHK	0xF5
ACK	0xF5	0x21	0	New baud rate ID	ACK_SUCCUSS ACK_FAIL	0	CHK	0xF5

Description:

Baud-rate IDs 1 through 5 correspond to 9600, 19200, 38400, 57600, and 115200 bps. The default after a system reset is 115200 bps.

Flag=0 changes the baud rate temporarily; power cycling restores its previous setting.

Flag=1 changes it permanently; the module stores the setting in nonvolatile memory, so it remains in effect after restart.

Example:

Send via serial port:

F5 21 00 01 00 00 20 F5

Set to 9600 bps.

The module checks whether the requested baud-rate ID is valid.

If valid, return success:

F5 21 00 01 00 00 20 F5

If invalid, return failure:

Example:

Send via serial port:

F5 21 00 07 00 00 26 F5

Invalid baud-rate ID; the module responds:

F5 21 00 07 01 00 27 F5

3.10 Acquire Image and Extract Features (8-byte CMD / >8-byte ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x23	0	0	0	0	CHK	0xF5

ACK data Format

i)Data head

Byte	1	2	3	4	5	6	7	8
ACK	0xF5	0x23	Hi(Len)	Low(Le n)	ACK_SUCCESS ACK_FAIL ACK_TIMEOUT ACK_BREAK	0	CHK	0xF5

ii) Data package

Byte	1	2	3	4	5...Len+1	Len+2	Len+3
ACK	0xF5	0	0	0	Feature Data	CHK	0xF5

Description:

Capture a fingerprint and extract its features. Total response length: 8206 bytes.

Example:

Host sends:

F5 23 00 00 00 00 23 F5

The module waits for the user to place a finger.

If no fingerprint is captured, the module times out after 8 seconds and returns:

F5 23 20 03 08 00 08 F5

If a finger is presented within the timeout, the module sends feature data as shown.

Note: This command also supports interruption. While waiting for a finger, the host may resend the command to exit capture mode. The module returns:

F5 23 20 03 18 00 18 F5

3.11 Get Image (8-byte CMD / >8-byte ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x24	0	0	0	0	CHK	0xF5

ACK data Format

i)Data head

Byte	1	2	3	4	5	6	7	8
ACK	0xF5	0x24	Width>>2	Height>>2	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

ii) Data package

Byte	1	2---Len+1	Len+2	Len+3
ACK	0xF5	Image Data	CHK	0xF5

Description:

Send via the serial port:

F5 24 00 00 00 00 24 F5

The module captures an image and transmits bitmap data on successful acquisition.

If no fingerprint is captured before timeout, it returns a timeout command.

F5 24 2C 2C 08 00 2C F5 (tested with a 176 x 176 sensor)

This command can also be interrupted. While waiting for a finger, the host may resend the current command to exit capture mode. The module returns:

F5 24 2C 2C 18 00 3C F5

Note: Tests used a 176 x 176 sensor.

3.12 Get Module Version Information (8-byte CMD / >8-byte ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x26	0	0	0	0	CHK	0xF5

ACK data Format

i)Data head

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x26	Hi(Len)	Low(Len)	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

ii)Data package

Byte	1	2...Len+1	Len+2	Len+3
ACK	0xF5	Version data	CHK	0xF5

Description:

Return module version data: firmware, sensor type, enrollment method, build time, etc.

3.13 Set/Get Matching Level (8-byte CMD/ACK)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x28	0	NEW matching level	0	0	CHK	0xF5
ACK	0xF5	0x28	0	Old matching level	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

Description:

Matching levels range from 0 to 2; default is 2. Higher levels are stricter.

The module restores the default on restart.

Matching levels and corresponding false acceptance rates are shown below:

Security level	False acceptance rate

0	1/10w
1	1/50w
2	1/100w

Example:

Set matching level 1 via serial port.

F5 28 00 01 00 00 29 F5

The module returns:

F5 28 00 01 00 00 29 F5 (previous matching level was 1)

3.14 Get Enrolled User IDs and Roles (8-byte CMD/ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x2B	0	0	0	0	CHK	0xF5

ACK data Format

i)Data Head

Byte	1	2	3	4	5	6	7	8
ACK	0xF5	0x2B	Hi(Len)	Low(Len)	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

ii)Data Package

Byte	1	2	3	4...Len+1	Len+2	Len+3
ACK	0xF5	User Number （higher 8bit）	User Number （lowe 8bit）	User ID and User Role	CHK	0xF5

Note: Len = 3 * User number + 2.

Format of user

Byte	4	5	6	7	8	9	…
Data	User ID1 （higher 8bit）	User ID1 （lower 8bit）	User Role1 （1/2/3）	User ID2 （higher 8bit）	User ID2 （lower 8bit）	User Role2 （1/2/3）	…

Example:

Send via serial port:

F5 2B 00 00 00 00 2B F5

If the database has no users, the module returns:

F5 2B 00 00 01 00 2A F5

If exactly one user is enrolled, it returns:

F5 2B 00 05 00 00 2E F5 F5 00 03 00 01 01 03 F5

3.15 Set/Get Enrollment Mode (8-byte CMD/ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x2D	0	0:Allowing same 1:Denying same	0:Set new enrollment mode 1:Get current enrollment mode	0	CHK	0xF5

ACK data Format

Byte	1	2	3	4	5	6	7	8
ACK	0xF5	0x2D	0x01	0:Allowing same 1:Denying same	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

Description:

There are two enrollment modes: allow duplicate and reject duplicate.

On power-up, the module defaults to allowing duplicate enrollment. Following power loss, it reverts to this default even if rejection had been enabled.

In reject-duplicate mode, after the first enrollment command, attempting to register an existing finger returns ACK_USER_EXIST (0x07).

ACK_USER_EXIST 0x07。

Example:

Set the mode to reject duplicate enrollment.

Send: F5 2D 00 01 00 00 2C F5 F5

Return: F5 2D 01 01 00 00 2D F5

3.16 Finger Detection (8-byte CMD/ACK)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x30	0	0	0	0	CHK	0xF5
ACK	0xF5	0x30	0	0	ACK_SUCCESS ACK_FAIL	0	CHK	0xF5

Description:

Immediately detect whether a finger is on the sensor: ACK_SUCCESS if present; otherwise ACK_FAIL.

3.17 Get Fingerprint Module Unique ID (8-byte CMD/ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x60	0	0	0	0	CHK	0xF5

ACK data Format

i)Data Head

Byte	1	2	3	4	5	6	7	8
ACK	0xF5	0x60	0	0	0	0	CHK	0xF5

ii)Data Package

Byte	9	10	11	....	17	18	19
ACK	0xF5	Fingerprint Module Unique ID Number				CHK	0xF5

Description:

Get the unique ID of the fingerprint module.

Send: F5 60 00 00 00 00 60 F5

Return: F5 60 00 00 00 00 60 F5 F5 38 30 32 30 39 04 2C 15 0E F5

3.18 Configure Enrollment (8-byte CMD/ACK)

CMD data Format

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0x3F	PID High	PID Low	PV	0	CHK	0xF5

Description:

PID PID High: high byte of parameter index.

PID PID Low: low byte of parameter index.

PV: parameter value.

Parameter index (PID) list:

0x0000: select 1CNR or NCNR enrollment mode.

0x0001: same-finger verification level.

0x0002: finger-lift detection configuration.

0x0003: number of fingerprint acquisitions, N.

0x0004: film spoofing detection setting.

0x0040: self-learning priority flag.

After power-up, the default is 3C3R enrollment with same-finger verification and without finger-lift detection. Other settings reset to defaults after power loss.

1. Select 1CNR / NCNR enrollment mode:

N is the number of fingerprint captures in either mode.

1CNR：F5 3F 00 00 05 00 3A F5

NCNR：F5 3F 00 00 06 00 39 F5

2. Same-finger verification level (command may be sent independently):

With 3C3R, at least one of captures 2 and 3 must match capture 1 for enrollment to succeed.

At level 0 the same-finger check is disabled; level 5 requires an exact match.

Level 0: F5 3F 00 01 00 00 3E F5

Level 1: F5 3F 00 01 01 00 3F F5

Level 2: F5 3F 00 01 02 00 3C F5

Level 3: F5 3F 00 01 03 00 3D F5

Level 4: F5 3F 00 01 04 00 3A F5

Level 5: F5 3F 00 01 05 00 3B F5

3. Finger-lift detection (command may be sent independently):

Enable finger-lift detection: F5 3F 00 02 01 00 3C F5

4. Set capture count:

3 captures: F5 3F 00 03 03 00 3F F5

4 captures: F5 3F 00 03 04 00 38 F5

5 captures: F5 3F 00 03 05 00 39 F5

5. Return first, then self-learn (command may be sent independently):

Send before matching: F5 3F 00 40 00 00 7F F5

Self-learning takes 150-250 ms. After receiving the matching response, wait at least 300 ms before power-off, or fingerprint self-learning may be damaged.

6. Film spoofing detection (off by default; command may be sent independently):

Before enrollment, send enable command: F5 3F 00 04 01 00 3A F5

Example 1: Set the enrollment capture count to 5 (1C5R / 5C5R).

1C5R: first send 1CNR mode command F5 3F 00 00 05 00 3A F5 (success response F5 3F 00 00 00 00 3F F5), then set capture count F5 3F 00 03 05 00 39 F5 (success response F5 3F 00 00 00 00 3F F5)。

5C5R: first send NCNR mode command F5 3F 00 00 06 00 39 F5 (success response F5 3F 00 00 00 00 3F F5), then set capture count F5 3F 00 03 05 00 39 F5 (success response F5 3F 00 00 00 00 3F F5)。

Two commands must be sent to set the enrollment count: first set 1CNR/NCNR mode, then set the capture count. This sequence is mandatory.

Example 2: Set same-finger verification level to 0.

Send: F5 3F 00 01 00 00 3E F5 (success: F5 3F 00 00 00 00 3F F5).

Example 3: Set six captures (NCNR), verification level 0 and finger-lift detection.

Send the following commands in order:

F5 3F 00 00 06 00 39 F5

F5 3F 00 01 00 00 3E F5

F5 3F 00 02 01 00 3C F5

F5 3F 00 03 06 00 3A F5

Send commands sequentially. Each success response is F5 3F 00 00 00 00 3F F5; wait for each success response before sending the next command.

3.19 LED Ring Control Command (8-byte CMD/ACK)

CMD data Format

i)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0xC3	led_Start	Led_End	Period	0	CHK	0xF5

ii)ACK data

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0xC3	0	0	0	0	CHK	0xF5

Description:

LED Start: initial color when switching between two LED colors.

LED End: final color when switching between two LED colors.

Period: time taken to change between colors, in 10 ms units. For Period=100, one transition takes 1 second. Valid range: 30-200; values outside this range cause the command not to execute.

Start and End are encoded in the lowest three bits of a uint8 value.

The lowest three bits represent Red, Green, Blue respectively (active low). Active-low outputs.

R G B

For example:

0x03 = 0x0000 0011: red LED on

0x05 = 0x0000 0101: green LED on

0x06 = 0x0000 0110: blue LED on

0x01 = 0x0000 0001: mixed red + green

0x02 = 0x0000 0010: mixed red + blue

Example 1:

Breathing LED

Red

F5 C3 03 07 96 00 51 F5

Green

F5 C3 05 07 96 00 57 F5

Blue

F5 C3 06 07 96 00 54 F5

Red + green

F5 C3 01 07 96 00 53 F5

Red + blue

F5 C3 02 07 96 00 50 F5

Green + blue

F5 C3 04 07 96 00 56 F5

Example 2:

Gradient LED

Red to blue

F5 C3 03 06 96 00 50 F5

Red to green:

F5 C3 03 05 96 00 53 F5

Green to blue:

F5 C3 05 06 96 00 56 F5

Red+green to red+blue:

F5 C3 01 02 96 00 56 F5

Red+green to green+blue:

F5 C3 01 04 96 00 50 F5

Red+blue to green+blue:

F5 C3 02 04 96 00 53 F5

Example 3: Solid-color LED

Red

F5 C3 03 03 96 00 55 F5

Green

F5 C3 05 05 96 00 55 F5

Blue

F5 C3 06 06 96 00 55 F5

Red + green

F5 C3 01 01 96 00 55 F5

Red + blue

F5 C3 02 02 96 00 55 F5

Green + blue

F5 C3 04 04 96 00 55 F5

Off (all LEDs off)

F5 C3 07 07 96 00 55 F5

3.20 Interrupt Command (8-byte CMD/ACK)

Byte	1	2	3	4	5	6	7	8
CMD	0xF5	0xFE	0	0	0	0	CHK	0xF5
ACK	0xF5	Type	0	0	ACK_BREAK ACK_FAIL	0	CHK	0xF5

Description: Immediately terminate fingerprint image acquisition.

Send during matching: F5 FE 00 00 00 00 FE F5

On successful interruption: F5 0C 00 00 18 00 14 F5

If idle: send F5 FE 00 00 00 00 FE F5; return F5 FE 00 00 01 00 FF F5

Revision History

Version Revision date Changes	Revised by	
V1.0	2019.12.31	Initial version	way
V1.1	2020.01.04	Revised	way
V1.2	2020.02.25	Added information on security levels	Xhaka
V1.3	2020.03.23	Added description of permanent baud-rate modification	Xhaka
V1.4	2020.03.24	Updated return values for 1:N matching	ZJ
V1.5	2020.03.26	Updated responses for certain commands	ZJ
V1.6	2020.04.18	Revised protocol for 8K-47.8 algorithm firmware	Xhaka
V1.7	2020.06.10	Added command 0x2B (3.14) and command 0xFE (3.20)	Xhaka